



INTRODUCTION TO SINGLE PAGE APPLICATION

WEBINAR

LET'S DESCRIBE YOURSELF



- ✓ What is SPA?
- ✓ I heard that before
- ✓ I'm working on that
- ✓ I'm a guru!

Traditional website

Evolution

Single Page Application

Challenges

Traditional Website



COMPONENTS

BEST FRIENDS

CSS



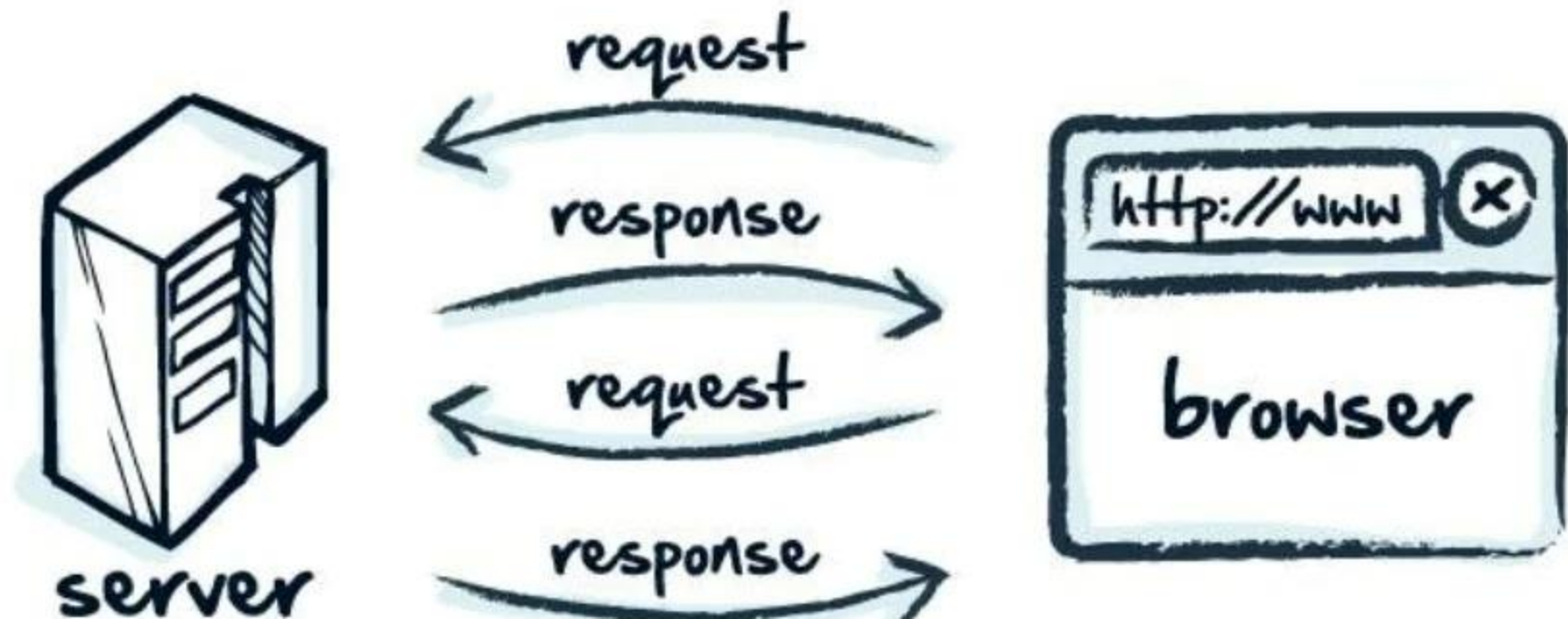
HTML



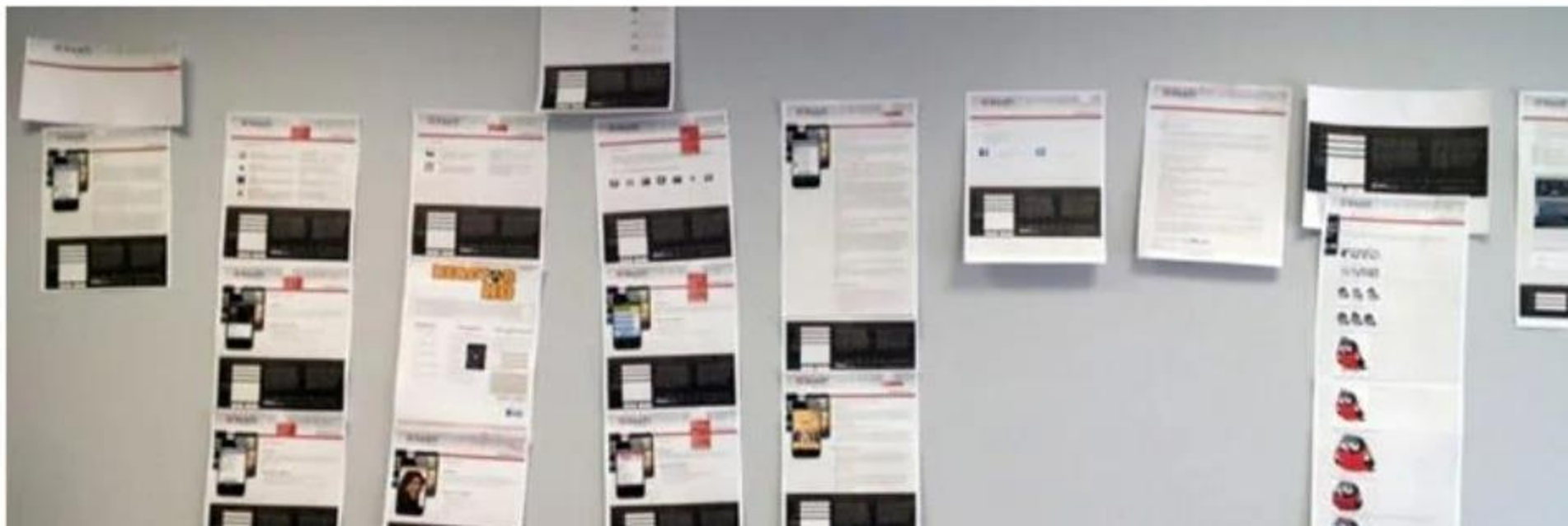
JS



SERVER-BASED APPLICATION



MULTIPLE PAGES

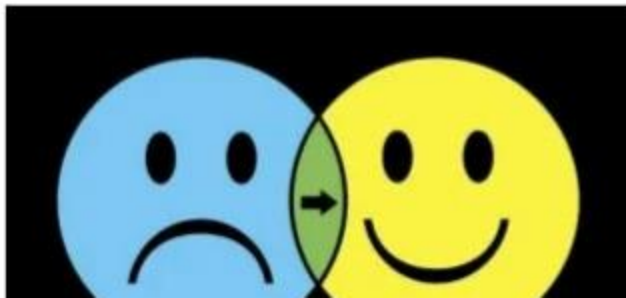


Evolution



NOWADAYS

- Modern browsers
- JavaScript becomes more powerful
- HTML5 - Thank God! You're here
- More focus on UX (User Experience)



CHANGES

- App-like user experience
- Faster UI, more interactive
- Be able to work Offline
- Support multiple platforms (responsive design)

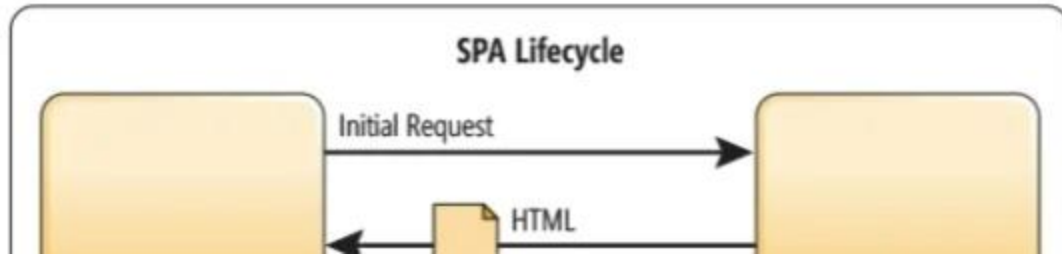
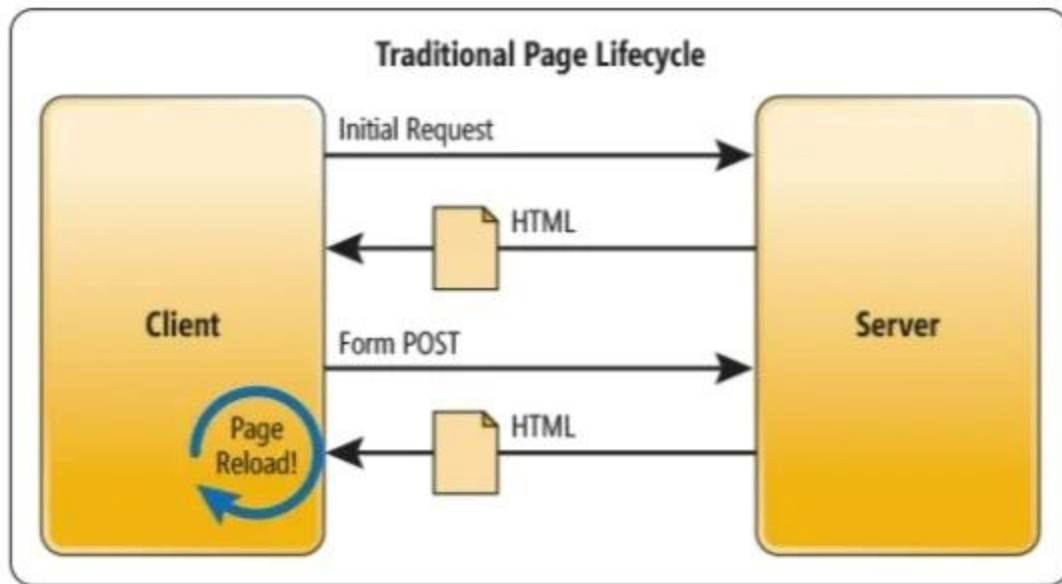
Single Page Application

SPA – SINGLE PAGE APPLICATION



- A next evolution from traditional website
- Web apps that load a single HTML page
- SPAs use AJAX and HTML5 to create fluid and responsive Web apps, without constant page reloads

SPA – SINGLE PAGE APPLICATION



BENEFITS



- No page flicker
- Javascript-based rendering
- Better UX
- Faster interaction

PRINCIPLES



- Performance matters
- Things you did on the server now live in the browser
- Render with JavaScript (library)
- Use CSS3
- Design with Offline in mind

-  Gmail
-  Google Maps
-  Airbnb
-  Netflix
-  Pinterest
-  Paypal
-  Many More.....

Examples

A large, solid blue arrow pointing to the left, positioned to the right of the "Examples" text.

HOW CAN I BUILD A SPA?

- Keep the principles in your mind
- Use a JavaScript framework adopted the principles



THAT'S IT???



Challenges



SEARCH ENGINE OPTIMIZATION



SEARCH ENGINE OPTIMIZATION SOLUTION



- Client side has to optimize the keywords, tags...
- Server side has to do 2 jobs:
 - Transfer data to client for SPA
 - Render whole website for Search Engine bot

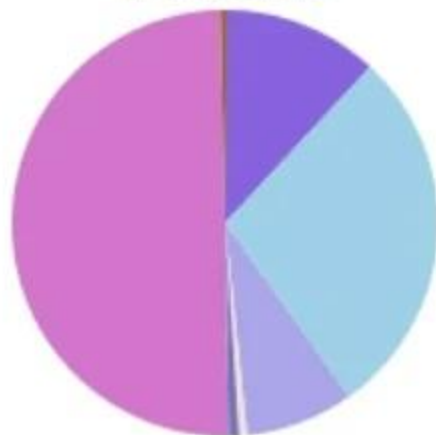


SPEED OF INITIAL LOAD

Grade **D** Overall performance score 61

F	Make fewer HTTP requests
F	Compress components with gzip
A	Put CSS at top
C	Put JavaScript at bottom
D	Avoid CSS expressions
C	Reduce DNS lookups
D	Minify JavaScript and CSS
B	Avoid URL redirects
A	Remove duplicate JavaScript and CSS
C	Reduce the number of DOM elements

Empty Cache



HTTP Requests - 88

Total Weight - 1012.0K

1	HTML/Text	120.7K
25	JavaScript File	285.1K
8	Stylesheet File	83.5K

SPEED OF INITIAL LOAD SOLUTION



BROWSER HISTORY



BROWSER HISTORY SOLUTION

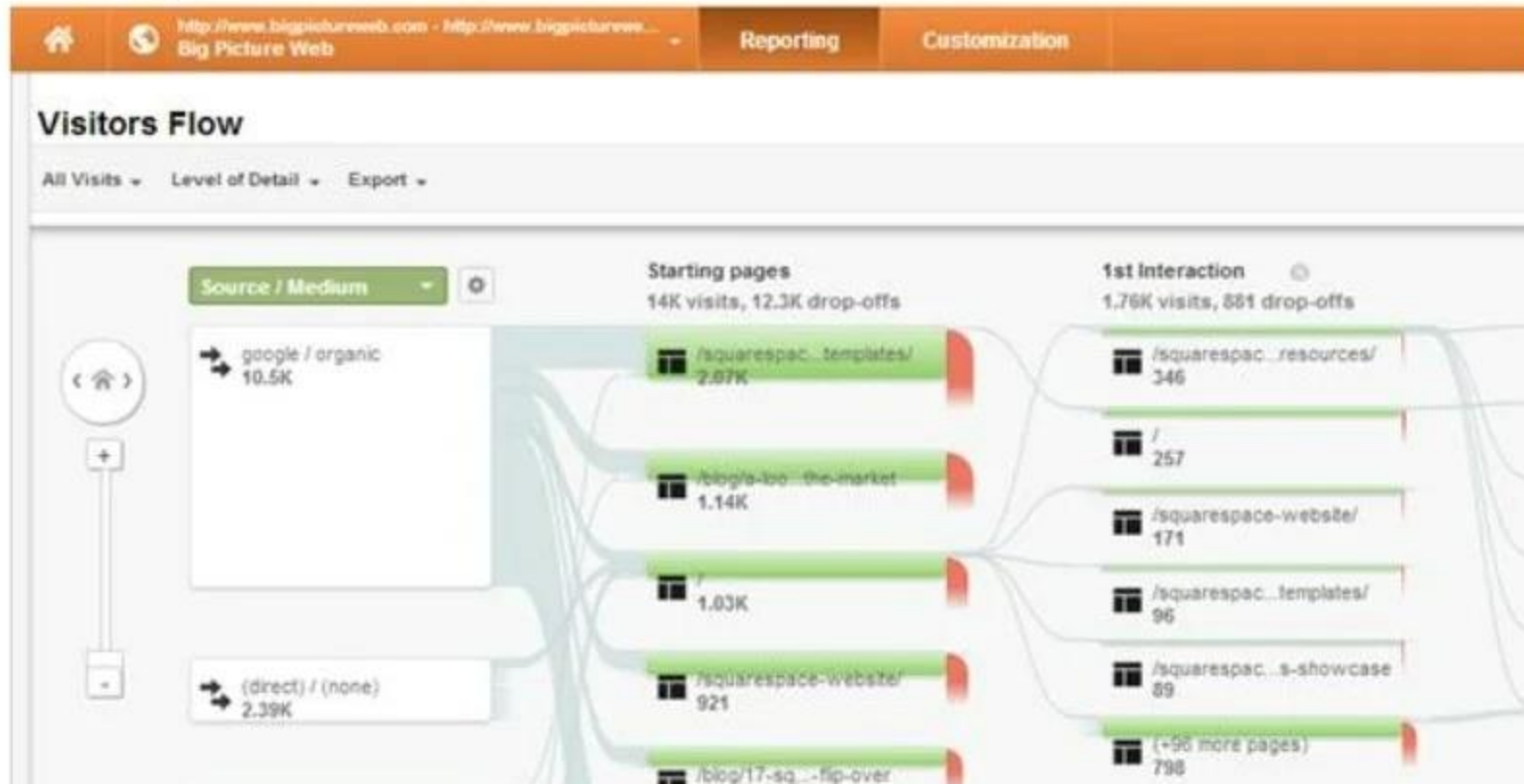
- Write JS code to handle the histories
- Use HTML5 pushState

```
function burger() {  
    // URL will change to /plate/burger  
    history.pushState( {  
        plate_id: 1,  
        plate: "Burger"
```

GOOGLE ANALYTICS



Google Analytics



GOOGLE ANALYTICS SOLUTION



CLIENT/SERVER CODE PARTITIONING



CLIENT/SERVER CODE PARTITIONING - SOLUTION

- Use a logic-less template language
- UI data binding



Takeaways



TAKEAWAYS

- Remind the traditional website
 - HTML, JS & CSS
 - Server-based application
- Introduction to SPA
- SPA principles
- Challenges

Q&A

Is my website suitable to apply SPA?

Thank You